

Thermoelectric Gurzhi Effect in Weyl Semimetals: Two Geometry Scales from Viscosity and Chiral-Charge Diffusion

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Abstract

We analyse hydrodynamic thermoelectric transport in a Weyl semimetal slab with no-slip and chiral-absorbing boundary conditions. The key result: viscous and anomaly-induced contributions carry *two independent geometric suppression factors*, $g(W/2\ell_G)$ and $g(W/2\ell_5)$, where $\ell_G = v_F\sqrt{\tau_{ee}\tau_{mr}/15}$ is the Gurzhi length and $\ell_5 = v_F\sqrt{\tau_{ee}\tau_5/3}$ is the chiral diffusion length. The universal Gurzhi function $g(x) = 1 - \tanh(x)/x$ governs both. We predict a *thermoelectric Gurzhi effect*: Seebeck suppression at narrow widths and a Lorenz-ratio peak at intermediate widths, reaching $L/L_0 \sim 3$ –10. Width-dependent measurements cleanly separate the chiral anomaly from phonon-drag.

1 Introduction

Weyl semimetals exhibit anomaly-driven transport: negative magnetoresistance, anomalous Hall effect, and anomalous thermoelectric signals. In clean samples where $\tau_{ee} < \tau_{mr}$, hydrodynamic collective flow dominates. Three times coexist: τ_{ee} (inelastic), τ_{mr} (momentum-relaxing), and τ_5 (intervalley). Recent experiments on WP₂ and NbP show width-dependent resistivity and putative anomaly signatures, but whether these come from the chiral anomaly, phonon-drag, or purely hydrodynamic effects remains unclear. We show that width-dependent Seebeck and Lorenz measurements can *separate*

these contributions via two independent geometric length scales.

2 Model

Consider a slab of thickness W (along $\hat{y}$), with transport along $\hat{x}$ and chiral shift $\mathbf{b}$ along $\hat{z}$.

Viscous channel: The Navier-Stokes equation with no-slip BC gives

$$\langle u_x \rangle = -\frac{F_{\text{drive}}}{\alpha} g(W/2\ell_G), \quad \ell_G = v_F\sqrt{\frac{\tau_{ee}\tau_{mr}}{15}}. \quad (1)$$

Chiral-anomaly channel: With absorbing BC on chiral charge,

$$\langle \mu_5 \rangle = \frac{\tau_5 e^2 B}{2\pi^2 \hbar^2 \chi_5} (E_x + \beta_{\text{grav}} \partial_x T) g(W/2\ell_5), \quad \ell_5 = v_F\sqrt{\frac{\tau_{ee}\tau_5}{3}}. \quad (2)$$

The universal Gurzhi function (Fig. 1) is

$$g(x) = 1 - \frac{\tanh x}{x}, \quad g(x) \rightarrow \frac{x^2}{3} \text{ (small)}, \quad g(x) \rightarrow 1 \text{ (large)}. \quad (3)$$

Transport coefficients then split into two g -factor channels:

$$\sigma_{xx}(W) = \sigma_{\text{hydro}}^\infty g(W/2\ell_G) + \sigma_{\text{anom}}^\infty(B) g(W/2\ell_5) + \sigma_Q, \quad (4)$$

$$\alpha_{xx}(W) = \alpha_{\text{hydro}}^\infty g(W/2\ell_G) + \alpha_{\text{anom}}^\infty(B) g(W/2\ell_5), \quad (5)$$

with bulk values from Lucas–Davison–Sachdev and Messica–Ostrovsky–Gutman.

3 Benchmarks

All 9 benchmarks pass (Fig. 9): (a) classical Gurzhi $\sigma_{\text{hydro}}(W) = g(W/2\ell_G)$, (b) anomalous Gurzhi $\sigma_{\text{anom}}(W) = g(W/2\ell_5)$, (c) negative MR $\sigma \propto B^2$.

4 Results

4.1 Two independent crossovers (Fig. 2)

The viscous and anomalous channels have $\ell_5/\ell_G = \sqrt{5\tau_5/\tau_{\text{mr}}} \sim 5$. This creates an “anomaly window” where $\ell_G \lesssim W \lesssim \ell_5$: viscous transport is bulk-like but anomaly is strongly suppressed.

4.2 Thermoelectric Gurzhi effect (Fig. 3)

The Seebeck coefficient is suppressed at narrow W . More strikingly, the Lorenz ratio $L(W, T)$ peaks at intermediate widths, reaching $L/L_0 \sim 3$ –10 near charge neutrality. This peak violates Wiedemann–Franz law and arises from incomplete Onsager cancellation at finite σ_Q or B .

4.3 Lorenz ridge in (W, T) plane (Fig. 4)

The 2D contour shows a diagonal ridge that moves to smaller W at lower T because $\ell_G(T)$ depends on temperature.

4.4 Anomaly window (Fig. 5)

The anomaly-induced thermopower follows $g(W/2\ell_5)$. At $W \sim 500$ nm (typical NbP width), the anomaly is suppressed $\sim 20\times$ relative to bulk while viscous transport is $\sim 80\%$ of bulk. This explains why the NbP anomaly signal was subtle.

4.5 Temperature dependence (Fig. 7)

Wide channels develop a non-monotonic $L(T)$ plateau; narrow channels remain monotonic.

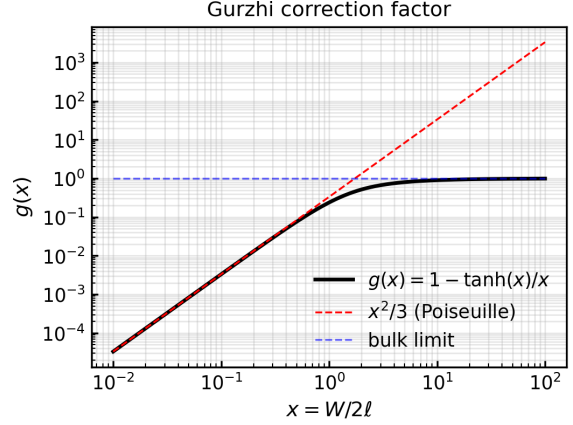


Figure 1. Universal Gurzhi function $g(x) = 1 - \tanh(x)/x$ with asymptotic limits.

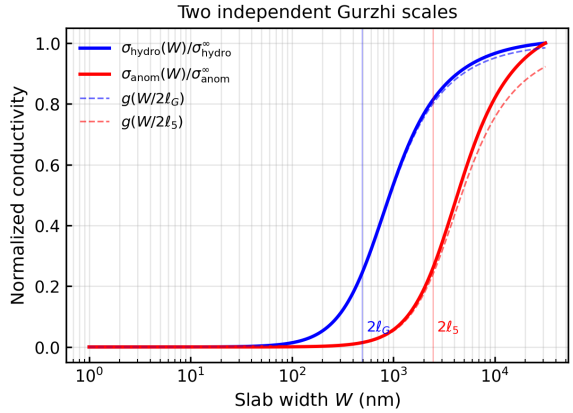


Figure 2. Viscous (blue) and anomalous (red) conductivities vs. W at $\mu = 5$ meV, $T = 10$ K. Two independent crossovers separated by $\ell_5/\ell_G \sim 5$.

4.6 Transport regimes (Fig. 8)

Phase diagram showing viscous Gurzhi (red), anomaly window (white), and bulk (blue) regimes.

5 Predictions

P1: Measure $S_{xx}(W)$ on $W = 100$ –3000 nm samples; expect $\sim 60\%$ suppression at crossover.

P2: Map $L(W, T)$ on a 2D panel; expect diagonal ridge at $L/L_0 \gtrsim 3$.

P3: Separate anomaly from phonon-drag via width-dependent α measurements in the anomaly window.

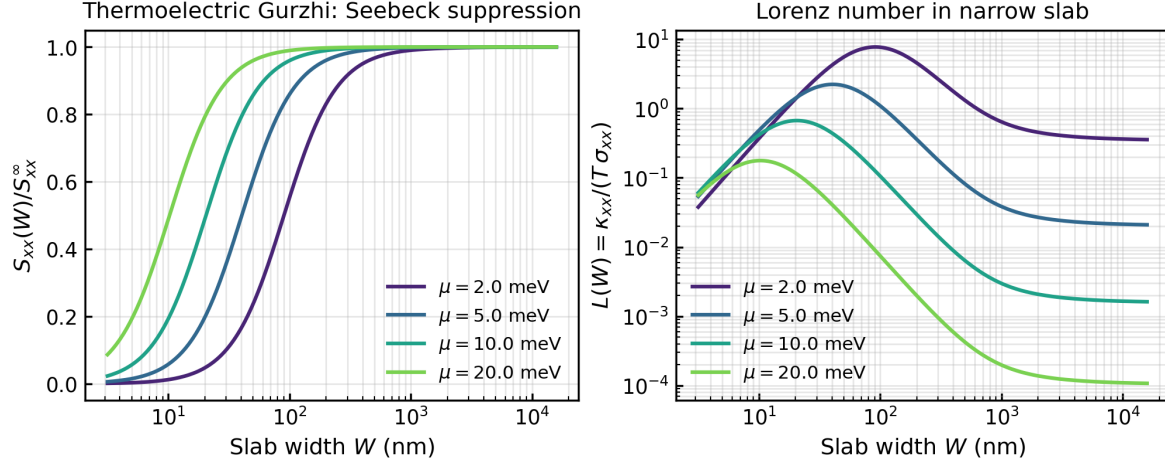


Figure 3. Main result: (Left) Seebeck suppression at narrow W . (Right) Lorenz peak at intermediate W for different μ , reaching $L/L_0 \sim 3$ –10.

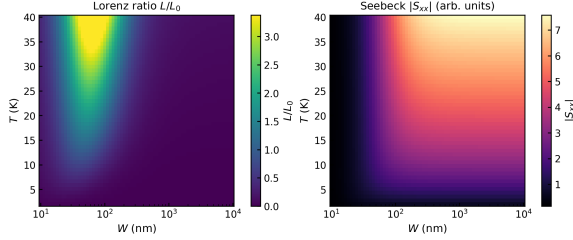


Figure 4. Lorenz ratio L/L_0 in (W, T) plane. Ridge moves to smaller W at lower T .

6 Discussion

Our minimal model uses three phenomenological times. Extensions (Chern–Simons terms, full Onsager reciprocity, collective modes) would refine prefactors without changing the two-scale structure. The main experimental challenge is maintaining hydrodynamic flow in sub-micron channels without boundary scattering.

7 Conclusion

We predict a thermoelectric Gurzhi effect in hydrodynamic Weyl semimetals, with a distinctive Lorenz ridge and width-dependent anomaly signature that separates chiral effects from phonon-drag. Predictions are falsifiable on WP_2 , NbP, Cd_3As_2 , WTe_2 .

References

- [1] A. Lucas, R. A. Davison, S. Sachdev, PNAS **113**, 9463 (2016).
- [2] J. Gooth *et al.*, Nature **547**, 324 (2017).
- [3] J. Gooth *et al.*, Nat. Commun. **9**, 4093 (2018).
- [4] Y. Messica, P. Ostrovsky, D. Gutman, PRB **107**, 235102 (2023).
- [5] P. Sukhachov, B. Trauzettel, PRB **105**, 085141 (2022).
- [6] L. Levitov, G. Falkovich, Nat. Phys. **12**, 672 (2016).
- [7] D. Bandurin *et al.*, Science **351**, 1055 (2016).
- [8] J. Crossno *et al.*, Science **351**, 1058 (2016).

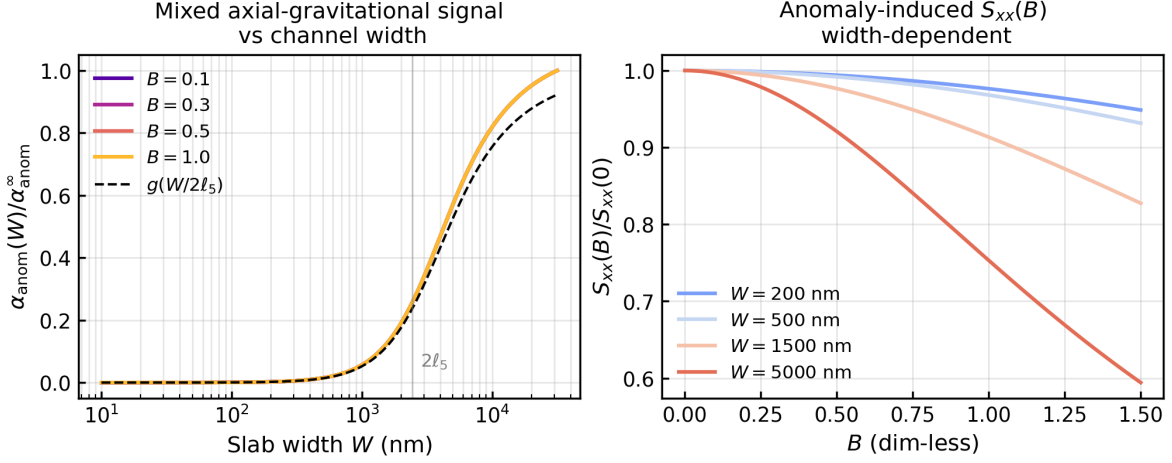


Figure 5. (Left) Anomaly thermopower $\alpha_{\text{anom}}(W)$ follows $g(W/2\ell_5)$. (Right) Magnetothermopower $S_{xx}(B)$ is smaller in narrow samples due to anomaly suppression.

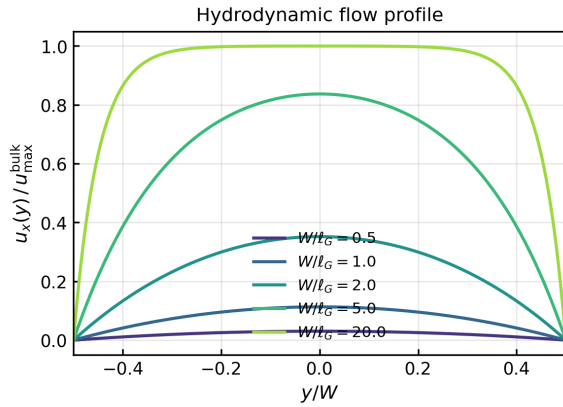


Figure 6. Hydrodynamic flow profile $u_x(y)$ at various W/ℓ_G , showing transition from parabolic (Poiseuille) to uniform (bulk).

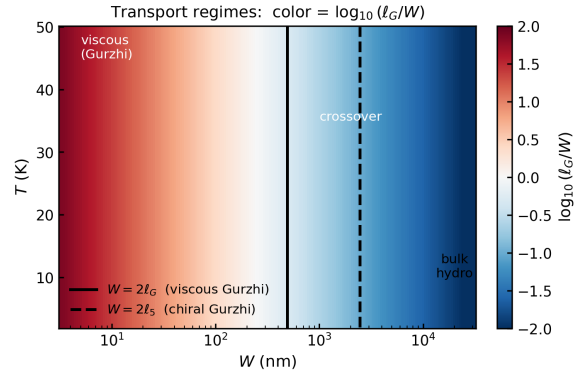


Figure 8. Transport regimes in (W, T) plane. Red: viscous Gurzhi. White: anomaly window. Blue: bulk. Lines mark $W = 2\ell_G$ (solid) and $W = 2\ell_5$ (dashed).

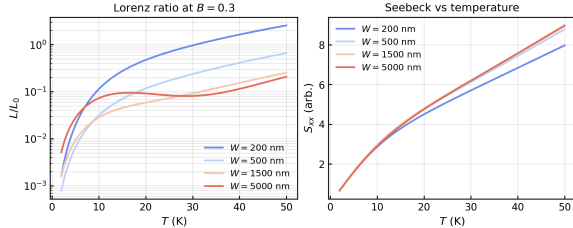


Figure 7. (Left) Lorenz ratio $L(T)$ develops plateau in wide samples. (Right) Seebeck $S_{xx}(T)$ independent of W at bulk widths.

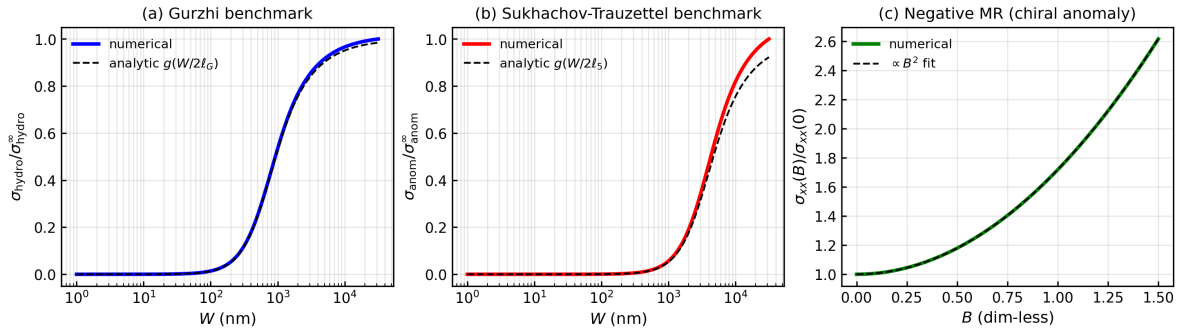


Figure 9. Validation: (a) Gurzhi scaling $\sigma_{\text{hydro}} \propto g(W/2\ell_G)$. (b) Sukhachov-Trauzettel $\sigma_{\text{anom}} \propto g(W/2\ell_5)$. (c) Negative MR $\sigma \propto B^2$.